

Build. Code. Create.

My robot workbook

Grades 6-8 | 8 lessons | 16 classes of 45 minutes

LEGO Education Computer Science & AI Kit 6–8 (#45522, 2026)



This picture shows an idea. Use LEGO's instructions to build the real robot.

Build a rover: a small robot that moves on wheels. Help it carry a paper parcel and stop at a red line. Try it, write down what happens, and make it better.

Before we start

Name: _____ Team: _____

Our robot's job

Start by carrying a paper parcel and stopping at a red line. Add one skill in each lesson. In the final lesson, choose one of five new jobs for the same robot base.

Work safely

Ask the teacher before each run. Start slowly. Keep one person ready at the Stop control. If the robot moves the wrong way, stop at once. If the screen control does not work, use the power-off method the teacher showed you. Test both ways to stop before students drive.

Keep fingers, hair and loose clothes away from the wheels. Turn the robot off before changing its parts. Stop if the sensor cannot name the color or the connection is lost. A red floor line tells this robot where to stop. It cannot use that color sensor to spot people, walls or table edges.

Before every drive

Clear the floor lane. Check that the parcel is secure and the sensor can see the floor. Keep Stop controls easy to reach. Say who will start and stop the robot. Ask the teacher to check your setup.

Keep useful notes

Write down every test, including ones that do not work. Change one thing at a time. Keep the same starting place, floor, parcel and code when you repeat a test.

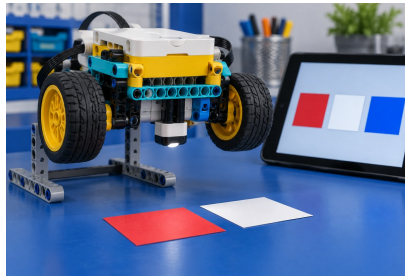
What do I know already?

Name a robot and describe its job:

How will our team stop the robot? Who will do each job?

Wake up your rover

Show a color to the sensor and explain the message that appears.



An imagined robot. Follow the official kit instructions.

Words to know

Sensor: A part that detects something, such as a color.

Program: The steps the robot follows.

Input / output: The color reading goes in. A screen message comes out.

What you need

The checked LEGO robot, its connected computer or tablet, and red, white and another color of paper.

Build and test

1. Ask your teacher to check the robot. Point to the sensor, wheels and computer. Keep the wheels off the floor.
2. Open the coding steps for lesson 1. Add no steps that move the wheels.
3. Make a rule: white shows READY; red shows DELIVERY; any other reading shows CHECK.
4. Try white, red and another color. Before each try, say which message you expect. Write down what appears.
5. Change one message to your team's own words. Try the colors again. Keep the wheels still.

What counts as success?

Each of the three colors gives the right message. The wheels do not move.

Open lesson 1's coding steps from the course page. Ask your teacher to check the program before running it.

Try it. Learn. Try again.

Name: _____ Date: _____

Saved code name: _____

What do I think will happen?

I think the robot will... because...

Make a fair test

Change the paper color. Keep the light and the sensor's position the same.

One thing I will change: _____

Draw the holder, starting setup or route

What happened?

Paper color	Message I expect	Message I see	Wheels still?
White			
Red			
Other			

What did I learn?

What is the input in this test? What is the output?

Next time I will change... because my test showed...

Build a color-test station

Name: _____ Team: _____ Date: _____

A wheeled robot base and a set of cards that make the screen respond.

Gather these materials

- [] LEGO set #45522 and its official instructions
- [] Connected computer or tablet
- [] Red, white and blue paper: one 8 cm square of each
- [] Two equal-height firm supports, checked by the teacher

Build: steps 1-4

- [] 1. Open LEGO's official instruction book. Find model C103 on pages 24-31. Build it with your teacher as a starting model, or use the wheeled base your teacher has already checked. The pictures in this course are project ideas, not exact LEGO assembly steps.
- [] 2. Ask the teacher to prepare a secure downward-facing color-sensor mount and clear attachment points for later lessons. Keep both wheel paths and all Stop controls clear. This is a change to the starting model and needs a real-kit test.
- [] 3. Set the robot on the two firm supports under fixed parts of its frame. Check that it cannot rock or slide. Both wheels must hang clear of the table. Do not support it by the sensor or an axle (a rod a wheel turns on).
- [] 4. Cut three paper squares, each 8 cm by 8 cm. Use red, white and blue. Put them beside the robot. Keep the same side of each paper facing the sensor.

[Open official LEGO instructions: C103, pages 24-31](#)

Before power: parts secure, wheels clear, Stop controls easy to reach.

Before a run: clear the lane and ask your teacher to check the setup.

After a run: stop and turn off the robot before changing any parts.

Drawings show the idea. Measure your own pieces; drawings are not full size.

Finish, code and test

Name: _____ Team: _____ Date: _____

Build: steps 5-8

[] 5. Connect only your team's motor and sensor in Coding Canvas. Point to the screen Stop control and the robot's power button. Ask the teacher to check the connection.

[] 6. Slide white paper below the sensor without touching it. Adjust the paper position until the sensor reads white. Mark that place with tape so every card goes in the same spot.

[] 7. Build the color-to-message program below. Keep all movement stopped. Test one card at a time, then put it back in the marked spot.

[] 8. Record all three results. Change one screen message to your team's words and test again. Leave the sensor mount and base ready for lesson 2.

Make the program

Start with all movement stopped.

Repeat 10 times: read the color. White shows READY; red shows DELIVERY; anything else shows CHECK. Wait 0.2 seconds before the next check.

Stop all movement and end. There are no driving steps in this program.

Teacher check: secure attachment, free wheels, clear sensor, reachable Stop controls, safe lane. Turn the power off before changing parts.

Record your color readings

Card color	Expected message	Actual message	Wheels still?
White			
Red			
Blue			

A successful build

The base is steady and both wheels stay still.

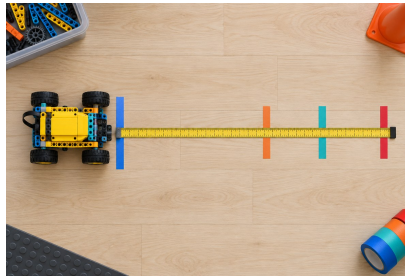
Each color produces the message in the rule.

Everyone can point to the sensor and the Stop controls.

How does the color make the screen message change?

Make a short drive

Make the robot stop after a short time. Measure how far it travels.



An imagined robot. Follow the official kit instructions.

Words to know

Motor: A part that makes something move.

Bounded: Set to stop after a certain time or number of steps.

Trial: One test run.

What you need

The checked robot, its connected device, a ruler and a clear floor lane.

Build and test

1. With the power off, check that both wheels can turn freely. Ask the teacher to show the Stop control and power-off method.
2. Use lesson 2's coding steps. Start with a low speed, such as 20 percent. Set the robot to move for 0.8 seconds, then stop.
3. Test with the wheels raised first. Then put the robot at the start line on the floor. Run it and measure the distance it travels.
4. Do three trials. A trial means one test run. Keep the same code and start line. Write down all three distances.
5. Ask the teacher to help you test the Stop control during a short drive. If the robot curves, check the wheels, floor and charge before changing one thing.

What counts as success?

The robot stops after each of three short drives. You also show that the Stop control works.

Open lesson 2's coding steps from the course page. Ask your teacher to check the program before running it.

Try it. Learn. Try again.

Name: _____ Date: _____

Saved code name: _____

What do I think will happen?

I think the robot will... because...

Make a fair test

Change only one thing, such as speed. Keep the time, floor, load and start line the same.

One thing I will change: _____

Draw the holder, starting setup or route

What happened?

Test number	Speed setting	Distance (cm)	Did it stop?
1			
2			
3			

What did I learn?

Why might the same code move the robot a different distance tomorrow?

Next time I will change... because my test showed...

Build a measured driving lane

Name: _____ Team: _____ Date: _____

A short, clear test lane for measuring the robot's travel.

Gather these materials

- ☐ Checked wheeled base and connected device
- ☐ Ruler or measuring tape
- ☐ Removable floor tape
- ☐ Paper and pencil

Build: steps 1-4

- ☐ 1. Choose a flat floor space with the teacher. Keep it away from stairs and table edges. Make the lane at least twice as wide as the robot.
- ☐ 2. Mark a start line across the lane. Put the ruler beside the lane, with zero at the start line. Keep the ruler out of the wheel path.
- ☐ 3. Add a small tape mark on the side of the robot, above its main wheel axle. Use this same mark to measure every run. An axle is the rod a wheel turns on.
- ☐ 4. With the power off, turn each wheel gently by hand. Remove anything rubbing a wheel. Check that the sensor mount stays clear of the floor.

Before power: parts secure, wheels clear, Stop controls easy to reach.

Before a run: clear the lane and ask your teacher to check the setup.

After a run: stop and turn off the robot before changing any parts.

Drawings show the idea. Measure your own pieces; drawings are not full size.

Finish, code and test

Name: _____ Team: _____ Date: _____

Build: steps 5-8

[] 5. Raise the wheels on firm supports. Try the short program below. If the wheels would send the robot backward, stop and ask the teacher to correct the motor direction.

[] 6. Place the robot's measuring mark over the start line. Clear the lane. With the teacher's approval, run the program and wait until the wheels stop.

[] 7. Measure from the start line to the same mark on the stopped robot. Write the distance in centimeters (cm). Return it to the start with the power off.

[] 8. Repeat three times with the same code. Then ask the teacher to help test the Stop control during another short run. Save the lane and measuring mark.

Make the program

Start with movement stopped. Set low speed, such as 20 percent.

Move forward for 0.8 seconds, then stop.

End the program. Do not use a forever-repeat block.

Teacher check: secure attachment, free wheels, clear sensor, reachable Stop controls, safe lane. Turn the power off before changing parts.

Record your test runs

Run	Speed setting	Distance (cm)	Stopped by itself?
1			
2			
3			

A successful build

The robot stops by itself after each short drive.

All three distances are recorded from the same mark.

The teacher checks the screen Stop control and power-off method.

Why might three runs travel slightly different distances?

Build a parcel holder

Build two versions of a holder and use your tests to choose one.



An imagined robot. Follow the official kit instructions.

Words to know

Cargo: The thing your robot carries.

Cradle: A holder that keeps the parcel in place.

Fair test: Change one thing and keep the rest the same.

What you need

The checked robot, a small paper parcel, light cardboard, easy-to-remove tape and teacher-approved spare LEGO pieces.

Build and test

1. Make one small paper parcel. Use the same parcel for every test. Draw a low holder that leaves the wheels, sensor and controls clear.
2. Build the holder. Ask the teacher to check it while the robot is turned off.
3. Use the same short drive from lesson 2. Try three runs. Write down whether the parcel stays on.
4. Change one part of the holder, such as the height of its sides. Keep the parcel and drive code the same. Try three more runs.
5. Choose a version using your test notes. Draw how it attaches, so another team can build it.

What counts as success?

Your holder keeps the parcel on in at least two of three runs. It does not rub a wheel or cover the sensor.

Open lesson 3's coding steps from the course page. Ask your teacher to check the program before running it.

Try it. Learn. Try again.

Name: _____ Date: _____

Saved code name: _____

What do I think will happen?

I think the robot will... because...

Make a fair test

Change one part of the holder. Keep the parcel, speed, time, floor and start line the same.

One thing I will change: _____

Draw the holder, starting setup or route

What happened?

Run / holder version	Parcel stayed on?	Anything rubbed?	What I noticed
1			
2			
3			

What did I learn?

Why should you keep the same parcel when you compare two holders?

Next time I will change... because my test showed...

Build two parcel holders

Name: _____ Team: _____ Date: _____

A low tray and an open-sided holder for the same paper parcel.

Gather these materials

- [] Checked base with teacher-approved fixed mounting points
- [] Thin cardboard: two rectangles about 12 cm by 8 cm
- [] Paper parcel about 4 cm by 3 cm by 2 cm
- [] Ruler, pencil, scissors and removable tape

Build: steps 1-4

- [] 1. Fold a small paper parcel about 4 cm long, 3 cm wide and 2 cm high. Tape it closed. Use this same light parcel for every test. Do not add coins or other weights.
- [] 2. Measure the clear space your teacher chose on the base. A 12 by 8 cm card is a starting size. Make it smaller if it would hang over a wheel or cover a control.
- [] 3. For holder A, draw a line 1 cm inside each edge of one rectangle. Cut out the four corner squares. Fold the four sides up along the lines and tape the corners. You have made a low tray.
- [] 4. For holder B, use the second rectangle. Fold its two long edges up by 1 cm and keep both short ends open. Put two folded paper stops near the short ends to stop the parcel sliding out.



Tray plan: cut away the four corner squares.

Fold up the four sides on the dashed lines.

Start with a 1 cm wall. Resize the base to fit.

Drawings show the idea. Measure your own pieces; drawings are not full size.

Finish, code and test

Name: _____ Team: _____ Date: _____

Build: steps 5-8

[] 5. Turn the robot off. Ask the teacher to show the mounting points. Fix holder A to those fixed points with the approved reusable connectors or paper tape tabs. Do not tape over the motor, sensor or buttons.

[] 6. Put the parcel in holder A. Raise the wheels and check for rubbing. Try the same short drive from lesson 2 three times and record whether the parcel stays on.

[] 7. Turn the power off. Swap to holder B. Use the same parcel, code, floor and start line for three more runs.

[] 8. Choose the better holder using your notes. Change only one feature if neither works well. Keep the selected holder for later lessons.

Make the program

Use lesson 2's checked program.

Low speed, 0.8 seconds forward, then stop and end.

Keep the same settings for both holders.

Teacher check: secure attachment, free wheels, clear sensor, reachable Stop controls, safe lane. Turn the power off before changing parts.

Record your test runs

Holder / run	Parcel stayed on?	Any rubbing?	What happened?
A / 1			
A / 2			
A / 3			
B / 1			
B / 2			
B / 3			

A successful build

The holder clears the wheels, sensor and controls.

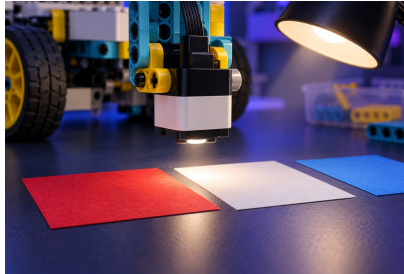
The parcel stays on in at least two of three runs.

The team records tests for both holders before choosing.

Which holder worked better? Use a result from your notes.

Help your robot read colors

Test paper colors and choose a setup that gives clear readings.



An imagined robot. Follow the official kit instructions.

Words to know

Reading: The information a sensor gives you.

Unknown reading: The sensor cannot name the color.

Fair test: Change one thing at a time.

What you need

The robot with its sensor looking down, red and white paper, another paper color, a ruler and your notes.

Build and test

1. Keep the wheels still. Check that the sensor looks down at the paper. Measure how high it is above the paper.
2. Open lesson 4's coding steps to show the color reading on the screen. Guess the reading before each try.
3. Try red three times, white three times and another color three times. Write every reading, including UNKNOWN.
4. Change only the sensor height or the room lighting. Try the same papers again. What changed?
5. Choose red and white paper the sensor can tell apart in your tests. Check them on the actual floor lane before driving.

What counts as success?

You record nine readings, try one change and explain which paper setup you will use.

Open lesson 4's coding steps from the course page. Ask your teacher to check the program before running it.

Try it. Learn. Try again.

Name: _____ Date: _____

Saved code name: _____

What do I think will happen?

I think the robot will... because...

Make a fair test

Change the light OR sensor height. Keep the other things the same.

One thing I will change: _____

Draw the holder, starting setup or route

What happened?

Paper / setup	Reading 1	Reading 2	Reading 3
White			
Red			
Other			

What did I learn?

The sensor names red correctly only twice in three tries. What should you do before driving?

Next time I will change... because my test showed...

Build a color-sensor test board

Name: _____ Team: _____ Date: _____

A board for comparing color readings at a fixed position.

Gather these materials

- [] Checked base with its downward-facing sensor
- [] Thin card about 30 cm by 12 cm
- [] Red, white and blue paper squares, each 8 cm wide
- [] Removable tape, ruler and pencil

Build: steps 1-4

- [] 1. Tape the three paper squares side by side on the card: red, white and blue. Leave a small gap between them. Keep the paper flat, without shiny tape over the reading area.
- [] 2. Keep the robot's wheels raised on firm supports. Slide the board under the sensor. Keep every motor stopped for this lesson.
- [] 3. Mark the board position that puts the center of red under the sensor. Do the same for white and blue. These marks help you line up each test.
- [] 4. Measure the gap from the sensor face to the board. Ask the teacher to check it against LEGO's guidance. Record the gap; do not assume one height works for every build.

Before power: parts secure, wheels clear, Stop controls easy to reach.

Before a run: clear the lane and ask your teacher to check the setup.

After a run: stop and turn off the robot before changing any parts.

Drawings show the idea. Measure your own pieces; drawings are not full size.

Finish, code and test

Name: _____ Team: _____ Date: _____

Build: steps 5-8

[] 5. Use the program below to show the color name. Test red three times, white three times and blue three times. Move the board away and back to its mark between readings.

[] 6. Write down all nine readings, even wrong or unknown ones. Unknown means the sensor cannot name the color.

[] 7. Change only one thing, such as room lighting or the sensor gap. Repeat all nine readings. Keep the paper and its position the same.

[] 8. Choose the setup that gives the most reliable red and white readings. Record the gap and lighting. Keep the same paper for the stopping lesson.

Make the program

Start with all movement stopped.

Repeat 10 times: read the color, show its name, and wait 0.2 seconds.

Stop all movement and end.

Teacher check: secure attachment, free wheels, clear sensor, reachable Stop controls, safe lane. Turn the power off before changing parts.

Record your color readings

Paper / setup	Reading 1	Reading 2	Reading 3
Red / first			
White / first			
Blue / first			
Red / changed			
White / changed			
Blue / changed			

A successful build

Nine readings are recorded for each setup.

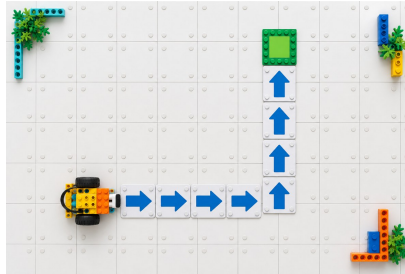
Only one thing changes in the comparison.

The team can explain the chosen red-and-white setup.

What did you change, and did it help the robot read colors?

Plan the delivery route

Put steps in order, then test and adjust a short route.



An imagined robot. Follow the official kit instructions.

Words to know

Sequence: Steps in order.

Calibrate: Test and adjust the settings.

Variable: One thing you can change, such as drive time.

What you need

The robot with an empty parcel holder, its connected device, floor tape and a ruler.

Build and test

1. Try a route in the screen practice area. Say where you think the robot will finish before pressing Run.
2. On the floor, mark a start, a turn point and a wide delivery space. Draw your route with arrows.
3. Use lesson 5's coding steps. Start with short, slow drives and a short turn. Test each part before joining them together.
4. Try the full route three times. Measure how far the robot stops from the middle of the delivery space.
5. Change just one drive time or turn time. Try three more runs. Use your notes to explain whether the change helped.

What counts as success?

The robot stops inside your marked space in at least two of three runs.

Open lesson 5's coding steps from the course page. Ask your teacher to check the program before running it.

Try it. Learn. Try again.

Name: _____ Date: _____

Saved code name: _____

What do I think will happen?

I think the robot will... because...

Make a fair test

Change one time setting. Keep the speed, floor, load and starting direction the same.

One thing I will change: _____

Draw the holder, starting setup or route

What happened?

Run / time setting	Distance from target	Turn too far / short?	Inside space?
1			
2			
3			

What did I learn?

Why does one good run not prove the route will work every time?

Next time I will change... because my test showed...

Build a route with arrow cards

Name: _____ Team: _____ Date: _____

An L-shaped route made from short drives and a turn.

Gather these materials

- ☐ Checked base, ruler and removable floor tape
- ☐ Six paper cards about 8 cm square
- ☐ Pencil or marker
- ☐ The short-drive code from lesson 2

Build: steps 1-4

- ☐ 1. Draw forward arrows on four cards, a right-turn arrow on one card and STOP on the last card. These cards are a plan; they do not control the robot by themselves.
- ☐ 2. Mark a short straight lane and a wide delivery space on the floor. Leave enough clear room for a turn. Keep arrow cards beside the lane, away from the wheels.
- ☐ 3. Arrange the cards: drive forward, turn right, drive forward, stop. Point to the direction the robot will face after each step.
- ☐ 4. Test the first short drive on its own. Adjust its time until it reaches the turning area. Calibrate means test and adjust a setting.

Before power: parts secure, wheels clear, Stop controls easy to reach.

Before a run: clear the lane and ask your teacher to check the setup.

After a run: stop and turn off the robot before changing any parts.

Drawings show the idea. Measure your own pieces; drawings are not full size.

Finish, code and test

Name: _____ Team: _____ Date: _____

Build: steps 5-8

[] 5. With the teacher, test a short, slow right turn by itself. Stop after the turn. Change its time a little if it turns too far or not far enough.

[] 6. Test the second short drive. Join the three checked parts, with a Stop after each one. Keep the first complete tests to 3 seconds of movement or less.

[] 7. Run the route three times from the same mark and direction. Record where the robot stops each time. Keep people and objects out of the lane.

[] 8. Change one time setting and try another set of runs. Save the code and draw the final route.

Make the program

Start stopped. Drive slowly for the tested first time, then stop.

Turn right slowly for the tested turn time, then stop.

Drive slowly for the tested second time, then stop and end.

Teacher check: secure attachment, free wheels, clear sensor, reachable Stop controls, safe lane. Turn the power off before changing parts.

Record your test runs

Run	First / turn / last times	Inside space?	What to adjust?
1			
2			
3			

A successful build

The program stops after each part and at the end.

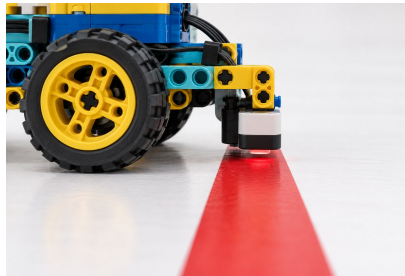
The robot finishes inside the space in at least two of three runs.

The team records the times and layout so it can try again.

Which time setting did you change, and what happened?

Stop at the red line

Make the robot move on white and stop on red or an unknown reading.



An imagined robot. Follow the official kit instructions.

Words to know

If-then rule: If this happens, do that.

Loop: Steps that repeat.

Bounded: Set to stop after a certain time or number of steps.

What you need

The checked robot, a white paper lane, a wide red finish strip, a ruler and a clear space past the strip.

Build and test

1. Use the paper and sensor setup you checked in lesson 4. Put a wide red strip across the white lane. Leave empty floor beyond it.
2. Use lesson 6's coding steps. Repeat no more than 20 times. Each time, read the color. On white, move slowly for 0.1 seconds, then stop. On red or any other reading, end the test.
3. Check the rule with the wheels raised. White should allow one short movement. Red and unknown must keep the wheels still.
4. Try the floor lane with the teacher at the controls. Measure where the sensor stops compared with the start of the red strip. If it misses the strip, record that too.
5. Do three runs. If the robot misses red, lower the speed OR make the strip wider. Try again. Do not let it restart by itself after a stop.

What counts as success?

Three runs stop when the sensor reads red. Unknown readings also stop the robot. It never restarts on its own.

Open lesson 6's coding steps from the course page. Ask your teacher to check the program before running it.

Try it. Learn. Try again.

Name: _____ Date: _____

Saved code name: _____

What do I think will happen?

I think the robot will... because...

Make a fair test

Change speed OR strip width. Keep the light, sensor height, start and white paper the same.

One thing I will change: _____

Draw the holder, starting setup or route

What happened?

Test number	Did it see red?	Distance past red line (cm)	What happened?
1			
2			
3			

What did I learn?

Why might the robot miss a narrow red strip even though it reads red while still?

Next time I will change... because my test showed...

Build a red-line stopping station

Name: _____ Team: _____ Date: _____

A white approach lane with a broad red finish strip.

Gather these materials

- ☐ Checked base with downward-facing color sensor
- ☐ The tested white and red paper from lesson 4
- ☐ Removable tape and ruler
- ☐ Connected device with lesson 6 code

Build: steps 1-4

- ☐ 1. Make a flat white lane at least twice as wide as the robot. Tape its edges down so the wheels cannot catch them. Keep tape away from the sensor's reading path.
- ☐ 2. Place a red strip across the full lane, starting about 10 cm deep in the direction of travel. This is a starting test size, not a guaranteed stopping distance.
- ☐ 3. Leave a large clear area beyond the red strip. Mark a start close enough that a short drive can reach it. Ask the teacher to check the whole layout.
- ☐ 4. With the wheels still, show the sensor white, red and another color. Fix wrong readings before driving. Keep the sensor gap from lesson 4.

Before power: parts secure, wheels clear, Stop controls easy to reach.

Before a run: clear the lane and ask your teacher to check the setup.

After a run: stop and turn off the robot before changing any parts.

Drawings show the idea. Measure your own pieces; drawings are not full size.

Finish, code and test

Name: _____ Team: _____ Date: _____

Build: steps 5-8

[] 5. Build the program below. A variable is a value the program remembers. keepGoing remembers whether this test may continue.

[] 6. At low speed, try one approach to red. Keep someone ready at Stop. When the program ends, measure how far the sensor moved past the start of the red strip.

[] 7. Repeat three approaches. Separately test an unknown or other-color reading with wheels raised. After a red or unknown reading, show white again: the same test must stay stopped.

[] 8. If it misses red, stop. Try a wider strip or a lower speed, one change at a time. Do not remove the 20-check limit or the Stop steps.

Make the program

Start stopped. Set keepGoing to Yes.

Repeat at most 20 times. Only if keepGoing is Yes: read the color.

If white: move slowly for 0.1 seconds, then stop. Otherwise: stop and set keepGoing to No.

After the repeats, stop and end. A person must start a new test.

Teacher check: secure attachment, free wheels, clear sensor, reachable Stop controls, safe lane. Turn the power off before changing parts.

Record your test runs

Run	Saw red?	Distance past line (cm)	Stayed stopped?
1			
2			
3			

A successful build

Three approach runs stop when red is read.

Red and unknown both end the test; white cannot restart it.

The program ends after no more than 20 checks.

What could make the robot miss a narrow red strip?

Make the delivery

Join your tested parts and check the whole delivery three times.



An imagined robot. Follow the official kit instructions.

Words to know

Success criteria: The things that must happen for a test to count as a success.

Version: One saved design or program before you change it.

Reliable: Works as expected again and again.

What you need

Your robot, tested parcel holder, paper parcel, white lane, red finish strip, ruler and test notes.

Build and test

1. Write the three success rules before you drive: parcel stays on; robot stops in the delivery space; robot hits nothing.
2. Mark the start and delivery space. Place the red strip using your stopping tests from lesson 6. Ask the teacher to check the setup.
3. Use your tested color-stop code and attach the parcel. Add a turn only after you have tested it on its own.
4. Run the same setup three times. Mark Yes or No for every rule. A run passes only when all three answers are Yes.
5. Choose one change from your notes. Give the new version a name. Try three more runs and compare the results.

What counts as success?

At least two of three runs meet all three rules. Keep a record of every run, including the ones that fail.

Open lesson 7's coding steps from the course page. Ask your teacher to check the program before running it.

Try it. Learn. Try again.

Name: _____ Date: _____

Saved code name: _____

What do I think will happen?

I think the robot will... because...

Make a fair test

Change one thing between versions. Keep everything the same during each set of three runs.

One thing I will change: _____

Draw the holder, starting setup or route

What happened?

Run / version	Parcel stayed on?	In space / no hit?	All three rules met?
1			
2			
3			

What did I learn?

Your first version passed once in three runs. Your new version passed twice. What can you say?

Next time I will change... because my test showed...

Build the complete delivery course

Name: _____ Team: _____ Date: _____

A robot that carries a parcel into a marked delivery space.

Gather these materials

- ☐ Checked base and the selected holder from lesson 3
- ☐ The same light paper parcel
- ☐ White lane and red strip from lesson 6
- ☐ Removable tape, ruler and saved code

Build: steps 1-4

- ☐ 1. Turn the robot off. Attach the tested holder to the approved mounting points. Put the paper parcel low in the center. Check that no part touches a wheel or covers the sensor.
- ☐ 2. Keep the white approach and red strip from lesson 6. Tape a delivery rectangle around the stopping area, large enough for the whole robot.
- ☐ 3. Keep the lane clear. For this test, use a short straight approach to the red strip. Add a longer route only after this works.
- ☐ 4. Use the checked color-stop program from lesson 6. Keep its short moves, Stop steps and maximum of 20 checks.

Before power: parts secure, wheels clear, Stop controls easy to reach.

Before a run: clear the lane and ask your teacher to check the setup.

After a run: stop and turn off the robot before changing any parts.

Drawings show the idea. Measure your own pieces; drawings are not full size.

Finish, code and test

Name: _____ Team: _____ Date: _____

Build: steps 5-8

[] 5. Write three rules before driving: parcel stays on; robot stops inside the delivery space; robot hits nothing. A run passes only when all three happen.

[] 6. Run the delivery three times from the same start. Write Yes or No for each rule after each run. Keep the failed runs in the notes.

[] 7. Choose one change, such as a lower holder wall or wider red strip. Keep the other parts and code the same. Try another set of three runs.

[] 8. Save the best-tested setup. Keep the base ready for the final build choice.

Make the program

Use the complete lesson 6 color-stop program.

White permits one 0.1-second slow move and then Stop. Red or any other reading ends the test.

Stop and end after at most 20 checks. Never restart within the same test.

Teacher check: secure attachment, free wheels, clear sensor, reachable Stop controls, safe lane. Turn the power off before changing parts.

Record your test runs

Run	Parcel stayed on?	Inside space?	Hit nothing?
1			
2			
3			

A successful build

At least two of three runs meet all three rules.

The robot still passes the red-and-unknown stop checks.

The team can name one change and the result it produced.

What improved when you changed one part of the delivery?

Choose a build and show it

Choose one of five builds, test it and explain how you improved it.



An imagined robot. Follow the official kit instructions.

Words to know

Attachment: A part you add to the robot for a new job.

Evidence: Measurements and notes that show what happened.

Handover: Directions that help someone else use your robot.

What you need

The same checked robot base, plus the materials listed for your chosen build.

Build and test

1. Open this module's build worksheet. Choose ONE of the five builds and read its two instruction pages.
2. Gather your choice's materials. Turn the power off, build the attachment and ask the teacher to check it. Use the program for that choice.
3. Try three runs with the same setup. Write down every result. Change one thing and test again. Save the final code with a clear name.
4. Ask another team to follow your setup, start and stop directions with the teacher nearby. Fix unclear instructions. Keep your recorded results unchanged.
5. Give a two-minute presentation. Explain the job and program, compare your results, and say what you would test next. Each student explains their own part.

What counts as success?

The chosen build meets its own rules in at least two of three runs. Another team can follow your directions. Explain one improvement and one remaining problem.

Open lesson 8's coding steps from the course page. Ask your teacher to check the program before running it.

Try it. Learn. Try again.

Name: _____ Date: _____

Saved code name: _____

What do I think will happen?

I think the robot will... because...

Make a fair test

Keep the same chosen attachment, test objects, code and start marks within each group of three runs.

One thing I will change: _____

Draw the holder, starting setup or route

What happened?

Final run	Chosen build	All its rules met?	What to try next
1			
2			
3			

What did I learn?

What should you do if your robot fails during the class demonstration?

Next time I will change... because my test showed...

Choose your robot's next job

Name: _____ Team: _____ Date: _____

Choose ONE build. Reuse the wheeled base you have already tested. Your teacher must approve its attachment points and each new setup before you run it.

Choice	What you build	Start in this packet
1. Parcel delivery robot	Carry a paper parcel to a red-line delivery space.	Pages 35-36
2. Drawing robot	Draw a short line on a protected paper surface.	Pages 37-38
3. Paper sweeper	Push three light paper pieces into a collection area.	Pages 39-40
4. Ball-pushing robot	Guide one light paper ball toward a goal.	Pages 41-42
5. Rescue carrier	Tow a light paper figure on a small rescue sled.	Pages 43-44

Our choice: _____

Why we chose it: _____

Before you build

☐ Our shared base passed the short-drive and Stop checks.

☐ We measured the clear mounting space.

☐ We found fixed attachment points with the teacher.

☐ We have the materials for one choice, not all five.

Mounting points means the places where you attach a new part. Use fixed parts of the frame. Do not fasten to wheels, axles, sensors, buttons or moving motor parts.

If the card sizes do not fit, make them smaller. Ask the teacher to approve the changes. These attachment plans are original teaching drafts and still need a real-kit tryout.

Plan, test and share

Name: _____ Team: _____ Date: _____

First class: choose and build

- ☐ Read your choice's two pages. Explain the robot's new job.
- ☐ Gather the materials. Build one step at a time with the power off.
- ☐ Check the new attachment with your teacher. Try one short run.

Second class: test and improve

- ☐ Use your choice's test table for three runs. Write every result.
- ☐ Change one thing and try again. Keep the original notes.
- ☐ Save the final code with a clear name.
- ☐ Ask another team to follow your setup, start and stop directions.

Shared check before every run

Teacher check: secure attachment, free wheels, clear sensor, reachable Stop controls, safe lane. Turn the power off before changing parts.

Show what you learned

Give a two-minute presentation: explain the job, show how the program works, compare results before and after a change, then name one next test.

My job on the team: _____

One choice I made: _____

A result that helped me decide: _____

A run passes only if it meets every rule for your chosen build. Aim for at least two passing runs out of three. All runs must keep the Stop controls available.

Choice 1: Parcel delivery robot

Name: _____ Team: _____ Date: _____

Carry a paper parcel to a red-line delivery space.

Gather these materials

- ☐ The shared checked base
- ☐ Thin card about 12 by 8 cm
- ☐ One paper parcel about 4 by 3 by 2 cm
- ☐ Ruler, scissors and removable tape



Tray plan: cut away the four corner squares.

Fold up the four sides on the dashed lines.

Start with a 1 cm wall. Resize the base to fit.

Measure your own pieces. The drawing is not full size. Resize any attachment to keep wheels and controls clear.

Build your attachment

- ☐ 1. Measure the clear mounting space on the base. Cut the card to fit it. Keep at least 1 cm of room between the attachment and any wheel. Make it smaller if needed.
- ☐ 2. Draw a fold line 1 cm inside each edge. Cut out the four corner squares. Fold all four walls up and tape the corners to make a shallow tray.
- ☐ 3. Fold a light paper parcel. Place it in the center of the tray. Add a small folded-paper stop at each end if it slides. Do not add heavy contents.
- ☐ 4. Turn the robot off. Use the teacher-approved mounting points and removable connectors to hold the tray flat. Keep buttons and the sensor clear.
- ☐ 5. Raise the wheels and check for rubbing. Then place the robot on the white lane from lesson 6. Keep the broad red strip and empty space beyond it.
- ☐ 6. Run the code below three times. If the parcel falls, change one tray feature and repeat. Unload the parcel by hand only after the robot has stopped.

Choice 1: program and tests

Name: _____ Team: _____ Date: _____

Make the program

1. Use lesson 6's tested color-stop program without removing its limits.
2. Start stopped; set keepGoing to Yes. Repeat at most 20 times. Only while keepGoing is Yes, read the color: white means a slow 0.1-second move then Stop; any other reading means Stop and set keepGoing to No.
3. Stop and end after the repeats. Red means delivery; unknown means check the setup.

Teacher check: secure attachment, free wheels, clear sensor, reachable Stop controls, safe lane. Turn the power off before changing parts.

What counts as success?

- ☐ The parcel stays in the tray.
- ☐ The whole robot stops inside the delivery space.
- ☐ The robot hits no object or person.

Keep the same setup for three runs

Run	Parcel stayed on?	Inside space?	Hit nothing?
1			
2			
3			

One change to try

Which tray feature helped the parcel stay on?

My change: _____

After changing one thing, try three more runs. Record them on the back of this sheet.

Explain your build

Show the job, the program, one result that helped you improve, and one thing that needs more work.

Choice 2: Drawing robot

Name: _____ Team: _____ Date: _____

Draw a short line on a protected paper surface.

Gather these materials

- [] The shared checked base
- [] One washable felt-tip marker
- [] A card strip about 12 by 3 cm and removable tape
- [] A large sheet of paper over a protective mat



Wrap this card around the marker.

Tape card to card to make a sleeve.

The marker tip should just touch the paper.

Measure your own pieces. The drawing is not full size. Resize any attachment to keep wheels and controls clear.

Build your attachment

- [] 1. Cover a large floor area with a protective mat and paper. Tape the paper edges flat. Ask the teacher to approve the marker and surface.
- [] 2. Wrap the card strip loosely around the marker to make a sleeve. Tape the card to itself. The marker must slide up and down inside it.
- [] 3. Turn the robot off. Fix the sleeve upright at a teacher-approved point behind the wheels. Keep it away from the sensor and moving parts.
- [] 4. Slide the uncapped marker down until its tip just touches the paper. Secure the marker in the sleeve with a small removable tape tab. It must not lift the wheels off the paper.
- [] 5. Raise the wheels and check the short program. Put the robot back on the protected paper with enough space for the whole run.
- [] 6. Draw three short lines. Stop and turn off the robot before lifting or adjusting the marker. If a line has gaps, lower the tip a little; if the robot struggles, raise it.

Choice 2: program and tests

Name: _____ Team: _____ Date: _____

Make the program

1. Start with movement stopped. Use a checked low speed, such as 20 percent.
2. Drive forward for 0.8 seconds, then stop and end. Shorten the time if the paper is too small.
3. No automatic repeat. Lift the marker and return the robot by hand between runs with the power off.

Teacher check: secure attachment, free wheels, clear sensor, reachable Stop controls, safe lane. Turn the power off before changing parts.

What counts as success?

- ☐ A visible line is drawn on the paper.
- ☐ The marker stays in its holder and all marks stay on protected paper.
- ☐ The robot stops by itself at the programmed time.

Keep the same setup for three runs

Run	Visible line?	Stayed on paper?	Stopped by itself?
1			
2			
3			

One change to try

How did marker height change the line or the robot's movement?

My change: _____

After changing one thing, try three more runs. Record them on the back of this sheet.

Explain your build

Show the job, the program, one result that helped you improve, and one thing that needs more work.

Choice 3: Paper sweeper

Name: _____ Team: _____ Date: _____

Push three light paper pieces into a collection area.

Gather these materials

- ☐ The shared checked base
- ☐ A thin card strip about 12 by 5 cm
- ☐ Three tissue-paper squares, each about 2 cm wide
- ☐ Removable tape, ruler and scissors



Fold the top strip back for mounting.

Cut slits from the bottom to make bristles.

Keep every slit below the fold line.

Measure your own pieces. The drawing is not full size. Resize any attachment to keep wheels and controls clear.

Build your attachment

- ☐ 1. Cut a card strip as wide as the robot, but no wider than the clear test lane. Fold its top 2 cm backward to make a mounting edge.
- ☐ 2. Cut short vertical slits about 1 cm apart into the lower 2 cm. These make flexible paper bristles. Round sharp card corners with scissors.
- ☐ 3. Turn the robot off. Attach the folded top edge to approved fixed front mounting points. The flexible ends should just brush the floor without lifting any wheel.
- ☐ 4. Use a clear floor lane. Mark a collection box farther along it. Place three tissue squares in a row across the starting sweep area. Keep them away from wheel paths.
- ☐ 5. Try the short program below. Check that the paper bristles can bend. Stop at once if a piece catches in a wheel; turn the power off before removing it.
- ☐ 6. Count the pieces that reach the box. Repeat three times with the same starting positions. Adjust the bristle height once and compare.

Choice 3: program and tests

Name: _____ Team: _____ Date: _____

Make the program

1. Start stopped and set the checked low speed.
2. Drive forward for 0.8 seconds, then stop and end. Set the collection box using the distance measured in lesson 2.
3. Do not add continuous driving. Return the paper pieces and robot to their marks between tests.

Teacher check: secure attachment, free wheels, clear sensor, reachable Stop controls, safe lane. Turn the power off before changing parts.

What counts as success?

- ☐ At least two of the three tissue squares reach the box.
- ☐ No paper gets caught in the wheels.
- ☐ The robot stops by itself and touches no object except the tissue squares.

Keep the same setup for three runs

Run	Pieces in box (0-3)	Wheels clear?	Stopped by itself?
1			
2			
3			

One change to try

Did shorter or longer bristles move the paper better?

My change: _____

After changing one thing, try three more runs. Record them on the back of this sheet.

Explain your build

Show the job, the program, one result that helped you improve, and one thing that needs more work.

Choice 4: Ball-pushing robot

Name: _____ Team: _____ Date: _____

Guide one light paper ball toward a goal.

Gather these materials

- ☐ The shared checked base
- ☐ Thin card about 14 by 5 cm
- ☐ One crumpled paper ball about 3 cm wide
- ☐ Removable tape and ruler



Top view: a shallow V guides the paper ball.

The opening faces forward.

Keep room for the ball to roll freely.

Measure your own pieces. The drawing is not full size. Resize any attachment to keep wheels and controls clear.

Build your attachment

- ☐ 1. Fold the card lengthwise so it has a 3 cm upright wall and a 2 cm mounting flap. Fold it gently across the middle to make a shallow V.
- ☐ 2. Make one light paper ball about 3 cm wide. Put it in the V opening. Leave enough room for it to roll freely; do not make a tight clamp.
- ☐ 3. Turn the robot off. Attach the two mounting flaps to teacher-approved fixed front points. Keep the V open at the front. Leave the wheels clear and keep the card just above the floor.
- ☐ 4. Keep the sensor clear even though this task uses a timed drive. Mark a wide goal across a short straight lane. Place the ball at the same start mark each time.
- ☐ 5. Use the short program below. The robot pushes the ball; it does not grip it. Stop if the ball rolls toward people or outside the clear test area.
- ☐ 6. Repeat three times. If the ball escapes sideways, adjust the V angle a little. Keep the same ball and code while comparing.

Choice 4: program and tests

Name: _____ Team: _____ Date: _____

Make the program

1. Start stopped and use the checked low speed.
2. Drive forward for 0.8 seconds, then stop and end. Put the goal within the tested travel distance.
3. No automatic repeat. Return the robot and ball to their marks with the power off.

Teacher check: secure attachment, free wheels, clear sensor, reachable Stop controls, safe lane. Turn the power off before changing parts.

What counts as success?

- ☐ The ball crosses the marked goal line.
- ☐ The guide stays attached and the wheels turn freely.
- ☐ The robot stops by itself and touches only the paper ball.

Keep the same setup for three runs

Run	Ball crossed goal?	Guide stayed on?	Stopped by itself?
1			
2			
3			

One change to try

Which V angle kept the ball in front of the robot?

My change: _____

After changing one thing, try three more runs. Record them on the back of this sheet.

Explain your build

Show the job, the program, one result that helped you improve, and one thing that needs more work.

Choice 5: Rescue carrier

Name: _____ Team: _____ Date: _____

Tow a light paper figure on a small rescue sled.

Gather these materials

- [] The shared checked base
- [] Thin card: one 12 by 6 cm piece and one 12 by 2 cm strip
- [] A flat paper person, two paper strips for bands, removable tape and scissors
- [] Ruler and a clear straight lane



Sled: fold the long sides up.

Wide tow bar: attach at the front.

Keep the bar away from the robot wheels.

Measure your own pieces. The drawing is not full size. Resize any attachment to keep wheels and controls clear.

Build your attachment

- [] 1. Use the 12 by 6 cm card as a sled. Fold each long edge up by 1 cm. Curl the front edge slightly upward so it does not catch on the floor.
- [] 2. Draw and cut out a small paper person. Lay it flat on the sled. Add two loose paper bands across the sled and tape their ends to the sides to hold the figure in place.
- [] 3. Use the 12 by 2 cm card strip as a wide tow bar. Tape one end flat to the sled's front. Make a small folded tab at the other end.
- [] 4. Before attaching it to the robot, pull the sled gently by hand using its wide tow bar. It should slide without catching. Keep it light and use only a straight route for this task.
- [] 5. Turn the robot off. Ask the teacher to connect the tow-bar tab to a fixed rear mounting point. Keep the whole strip centered behind the robot and away from the wheels. Do not use loose string.
- [] 6. Run the short program below three times. If the sled catches or a wheel struggles, stop. Smooth the sled edge or shorten the route, then test again.

Choice 5: program and tests

Name: _____ Team: _____ Date: _____

Make the program

1. Start stopped and use the checked low speed.
2. Drive straight for 0.8 seconds, then stop and end. Do not add turns or reversing for this first sled build.
3. Return the robot and sled to the start with the power off. Do not move a real person, animal or heavy object.

Teacher check: secure attachment, free wheels, clear sensor, reachable Stop controls, safe lane. Turn the power off before changing parts.

What counts as success?

- ☐ The paper person stays on the sled.
- ☐ The tow bar stays attached and away from the wheels.
- ☐ The robot moves the sled and stops by itself.

Keep the same setup for three runs

Run	Person stayed on?	Tow bar clear?	Moved and stopped?
1			
2			
3			

One change to try

What helped the sled slide without catching?

My change: _____

After changing one thing, try three more runs. Record them on the back of this sheet.

Explain your build

Show the job, the program, one result that helped you improve, and one thing that needs more work.

Show another team

Chosen robot job: _____

Saved code name: _____

Our rules for success

Draw the test area and start position

Setup, start, stop and try again

My two-minute presentation

Explain the job. Show the program. Compare test results before and after a change. Say what you would test next.

Another team's feedback

An instruction that was clear: _____

An instruction to improve: _____

My part in the build

Word help

A glossary is a list of words and their meanings. Use this page when you meet a word you do not know.

Word	Plain English meaning
Rover	A small robot that moves on wheels.
Program	A list of steps a computer or robot follows.
Coding block	A piece on the screen that gives the robot an instruction.
Sensor	A part that detects something, such as a color.
Motor	A part that uses electricity to make something move.
Input / output	Input is information going in. Output is the action or message that comes out.
Bounded	Set to stop after a certain time or number of steps.
Trial	One test run.
Fair test	Change one thing. Keep the other things the same.
Cargo / cradle	Cargo is what the robot carries. A cradle is a holder that keeps it in place.
Calibrate	Test and adjust settings so the robot does what you need.
Sequence / loop	A sequence is steps in order. A loop repeats steps.
Variable	One thing that can change, such as speed or time.
Mean	An average: add the numbers, then divide by how many numbers you added.
Success criteria	The things that must happen for a test to count as a success.
Evidence	Measurements or notes that show what happened.
Attachment	A part you add to the robot for a new job.

What will the kits cost?

These are US prices for LEGO Education Computer Science & AI 6-8 (#45522). The supply amounts are estimates: money set aside for materials.

What to buy	US dollars	What the amount covers
One kit · up to 4 learners	\$529.95	LEGO's listed price
Six individual kits · 24 learners	\$3,179.70	6 × \$529.95; fewer charging extras than bundle
Classroom bundle · 24 learners	\$3,499.00	6 kits plus charging and replacement extras
Supplies for a small tryout	\$35.00	Money set aside for: paper, tape, ruler and cargo materials
Supplies for six teams	\$120.00	Money set aside for six teams
Small tryout total before tax / shipping	\$564.95	1 kit + \$35 supplies; using a device you already have
Class total before tax / shipping	\$3,619.00	1 bundle + \$120 supplies; using 6 devices you already have

Prices were checked on LEGO's US website. Check them again before buying. The totals do not include tax, shipping, computers, school computer support or teacher training. The course and screen practice area do not need a paid subscription. No kit has been bought.

What comes with the class bundle?

The bundle has six kits, three extra multi-chargers, eight extra connection cards and a replacement brick pack. Buying six single kits costs \$319.30 less, but does not include all those extras.

Start small

Try one kit with four students and a computer or tablet you already have. For 24 students in teams of four, use six kits. For pairs, 12 kits cost \$6,359.40 before other costs. One spare kit adds \$529.95. These are kit purchases, not monthly fees.

Buying an older kit?

EV3 and Robot Inventor use different parts and coding apps. These lessons' coding steps would need changes. SPIKE direct sales ended June 30, 2026; LEGO says its app will be supported through June 30, 2031. Before buying used, check that the parts, charger and app you need are available.

[Selected kit and current US pricing](#)

[SPIKE retirement and support](#)